

Review and Comparative Study of Bi-Directional DC-DC Converters

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Abstract— The bi-directional DC-DC converters are utilized in numerous applications based on their both directions power transfer capability. This paper aims to discuss an in-depth literature review and comparative analysis of the various kinds of the bidirectional dc-dc converter. In this paper, each converter is classified according to the characteristics, structure, and voltage boosting techniques. A broad, comprehensive study and complete assessment have been done in this paper to select the converter's suitable applications. Moreover, this study helps to optimize the topologies in the future. Furthermore, the detailed gain comparison analysis of each non-isolated converters is presented. Finally, a detailed summary of each converter is explained with the merits, demerits, power densities, features, efficiencies, and components.

Keywords— *bidirectional, dc-dc converters, high voltage gain, battery, energy storage systems integration, efficiency, power density*

I. INTRODUCTION

Bi-directional dc-dc converters gradually became a more prominent research area in the field of power-electronics because it has an extensive range of applications such as wind farms [1], electric vehicle (EV) [2], aircraft applications, UPS, solar farms, X-ray, and military applications, etc [3-38]. These converters are operated at a diverse range of voltages, higher efficiency, and capable of transferring power in both ways. Bi-directional dc-dc topologies are also used in smart grids and electric vehicle charging stations [2] and [38]. In-vehicle to grid technology, the bidirectional converter is employed to charge the battery, and (vice versa) the energy in batteries can send back to the grid side. The bidirectional dc-dc converter helps to save energy while regenerative braking takes place in an electric vehicle.

Recent trends in the aviation industry suggest that mechanical systems resembling hydraulic and pneumatic which are powered by fuels in aircraft, cargo planes, and aerial vehicles, were replaced by electric counterparts like motor drives, bi-directional, and high gain step-up dc-dc converters[27]. Fossil fuel stations are replaced with electric power distribution systems for low maintenance. In electric power distribution systems, bi-directional converters have a vital role in increasing efficiency and reducing the number of components. Bidirectional dc-dc converters are used to improve the power distribution system's [34] efficiency and the grid's reliability. These converters are also used in HVDC transmission, which performs many functions other than regulating voltage, such as controlling the power flow and provides isolation during the faults. A bidirectional dc-dc converter is used in the hybrid DC distribution system to integrate renewable energy [6] sources like photovoltaic (PV)

modules, wind energy, etc. This application is used to charge the battery during low demand times and supply the battery's required additional power during peak load time. A bidirectional dc-dc converter is employed to charge a mobile from a power bank and sends the power back to the power bank in a mobile charging application. Besides, it also provides stabilization of voltage.

A hybrid energy source system [10], [13] with super capacitors and batteries is used in EV applications. It supports to improving the efficiency and battery life of EVs. The hybrid energy source provides the necessary power whenever a sudden change in acceleration occurs. The terminal-voltage of super capacitors is lower, and to meet the above requirement, a bidirectional dc-dc converter with a wide range of voltage gain with good dynamic behaviour is required. The converter requires galvanic isolation, high voltage dc gain, and dynamic behaviour for the above-mentioned applications. Each converter is suited for specific applications based on its features, merits, and demerits. This paper discusses the in-depth review of each converter based most demanded factors.

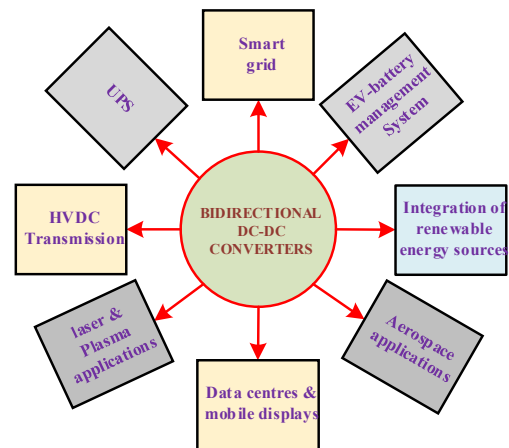


Figure-1.various applications of bidirectional dc-dc converter

II. CLASSIFICATION OF BI-DIRECTIONAL DC-DC CONVERTERS

The bidirectional converters are classified into two groups such as nonisolated and isolated. Further, each group is divided into subcategories according to the voltage boosting techniques, characteristics, and configuration. The various demanded parameters such as merits, demerits, features, power density, and complexity are discussed in detail. The step-up and step-down voltage gain ratios of different non-isolated and isolated topologies are depends on the turn's ratio of inductors and transformer and passive elements.

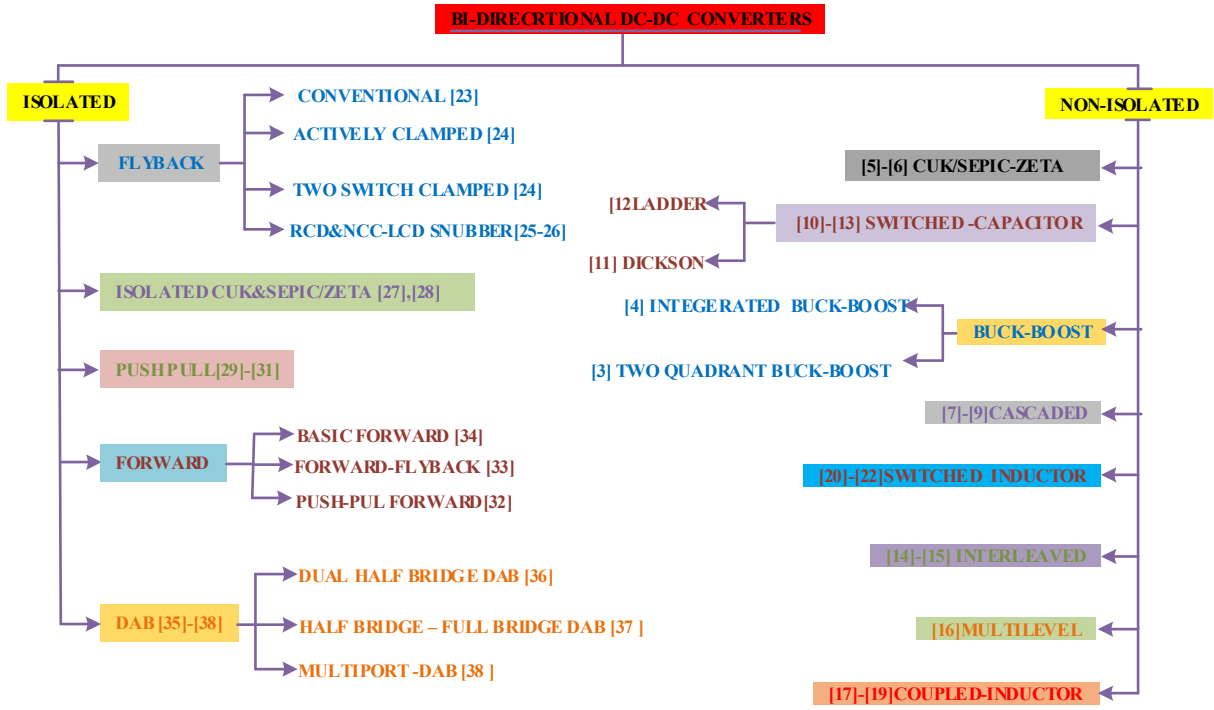


Fig-2.classification of bi-directional dc-dc converter

III. COMPARATIVE SURVEY

A. Non- Isolated Bidirectional DC-DC Converters

These converters have a simpler configuration due to non-utilization of the transformer; hence the weight, size, and cost reduction, which makes it more suitable to the aerospace application for power backup. Traditionally, Non-isolated bi-directional topologies can be derived by adding a diode with a controllable switch and vice versa. Some of the non-isolated bidirectional converters are described below.

1) Buck-Boost

The Buck-boost bi-directional dc-dc converter is derived from the conventional unidirectional buck and boost converter. The buck-boost has been divided into two categories [3-4]. Two quadrant buck-boost converters [3] are a modification of a single quadrant chopper with additional components, which makes the input current continuous. These converters are frequently used in dc motor drive and charging applications. In [4], the integrated buck-boost converter is basically configured with a bidirectional switch replaced by the unidirectional switch to flow the power in both directions. However, buck-boost converters provide a negative output voltage. Therefore, it is not suitable for most applications.

buck-boost	N_S	N_L	N_C	K	$\eta\%$	
					up	down
[3]	4	2	3	$\frac{-D}{1-D}$	96	96
[4]	2	2	2	$\frac{-D}{1-D}$	94.3	94.3

* N_S -number of semiconductors, N_L - number of inductors, N_C -number of capacitors, K-voltage gain

2) CUK&SEPIC-ZETA

The key feature of using the Cuk converter [6] is continuity in input and output currents. Further development of cuk converter with coupled inductor topology leads to

nullify the output and input current ripples. Cuk converter also provides the positive output voltage.

Sepic-zeta [5] converter reduces the input current ripples, and power semi conductive components of sepic-zeta [5] can be operated at zero current switching(ZCS) switched ON and OFF by employing soft-switching commutation. Due to its cascaded configurations, the efficiency is low when compared to other converters

Cuk/zeta-sepic	N_S	N_L	N_C	K	$\eta\%$	
					up	down
[5]	4	3	3	$\frac{1-D_2}{1-D_1}$	96	96
[6]	2	3	4	$\frac{D}{1-D}$	92	92

3) Switched Capacitor

switched capacitor	N_S	N_L	N_C	K		$\eta\%$	
				up	down	up	down
[10]	4	1	4	$\frac{2}{(1-D)}$	$\frac{D}{2}$	94.39	94.45
[11]	5	2	6	$\frac{2+D}{1-D}$	$\frac{D}{3-D}$	94.09	94.41
[12]	5	3	4	$\frac{1-\frac{D}{3}}{(1-D)^2}$	$1.5D^2$	95.5	97.5
[13]	4+2	2	1	$\frac{1}{(1-D)^2}$	D^2	88.7	88.7

The switched-capacitor [10-13] converters have a simple structure, a high voltage ratio. However, it has drawbacks of high voltage stresses across switches and capacitors. The converter proposed in [10,13] has fewer components with a better voltage gain ratio. A switched-capacitor has divided into two categories, such as Dickson and ladder configuration based [12] on the structure. These topologies are mainly used for high voltage gain. The switched capacitor with resonant inductor-based topology and coupled inductor topologies are derived in [11]. This converter improves the DC voltage-

gain and attains soft-switching and soft-charging with reduced voltage spikes

4) Cascaded DC-DC converters

Cascaded dc-dc converters [7-9] are mainly used for voltage boost ability, and apart from the high voltage gain, it has some drawbacks of higher switching-stress and less efficiency. In [8, 9], converters have the advantages of providing continuous current input/output sides and reducing the ripples in charging and discharging currents. These converters require more number of components and complexity in the controller.

cascaded	N_s	N_L	N_C	K		$\eta\%$	
				up	down	up	down
[7]	4	2	1	$\frac{1}{(1-D)^2}$	D^2	97.2	96.4
[8]	6	2	5	$\frac{3-D}{(1-D)^2}$	$\frac{D^2}{2+D}$	95.7	96.2
[9]	4	2	1	$\frac{1}{(1-D)^2}$	D^2	94.09	94.41

5) Interleaved

The interleaved converter is proposed in [14] reduces the current ripples on the low voltage side (LV). Further, the series-inductor connection is developed on the high-voltage (HV) side to attain the wide range of voltage gain, better dynamic performance, and thermal cooling system. It eliminates the switching frequency harmonics by multiplying fundamental frequency with several phases. Similarly, the conventional two-phase interleaved bi-directional converter in [15] reduces the harmonics on the lower voltage-side; however, it has demerits of high voltage stress across power switches and requires more number components.

Interleaved	N_s	N_L	N_C	K		$\eta\%$	
				up	down	up	down
[14]	5	2	3	$\frac{2}{(1-D)}$	$\frac{D}{2}$	95.21	95.30
[15]	8	4	3	$\frac{4}{(1-D)}$	$\frac{D}{4}$	96.4	94.5

6) Multilevel Converter

The multilevel bidirectional dc-dc converters are generally designed for high power applications and divided into two categories; those are modular and cascaded. These converters are [16] compact and less costly due to no inductors utilized in the circuits. A new cascaded multilevel proposed in [16] is used to reduce the voltage stress across the power switches; besides, it requires more switches, and it has control complexity.

multilevel	N_s	N_L	N_C	K		$\eta\%$	
				up	down	up	down
[16]	15	0	6	$\frac{N}{(1-D)}$	$\frac{D}{N}$	91.8	91.9

7) Coupled-Inductors

Coupled-inductors-based topologies in [18] are used to attain a wide range of gain. Here, a tapped series capacitor is proposed to achieve the soft switching. It is a further extension of coupled inductors reported in [17]. An extra capacitor with a tapped series inductor is used to form a resonant tank, which helps achieve soft-switching and improves the voltage withstanding capability. The

interleaved coupled and parallel inductor topologies reported in [17, 19] decrease the ripples at input-current by sharing the current. A tapped series capacitor with the interleaved method can reduce the switching-stress across the power switch. A coupled inductor with clamped circuit cells and voltage multipliers can improve soft switching and high-voltage gain with higher efficiency.

coupled inductor	N_s	N_L	N_C	K		$\eta\%$	
				up	down	up	down
[17]	4	2	3	$\frac{1+N}{(1-D)} + N$	$\frac{D}{1+N+ND}$	97.33	96.23
[18]	6	2	2	$\frac{2+ND}{(1-D)}$	$\frac{D(1-D)}{1+2N(1-D)^2}$	96.5	96.3
[19]	3	1	1	$\frac{1+D}{(1-D)}$	$\frac{D}{(2-D)}$	92.7	93.7

8) Switched-Inductor

Switched-inductor (SI) proposed in [20] and [21] have voltage boost capacity, and it is appropriate for medium power applications due to less current ripples and diminish input pulsating current issues. A new hybrid topology is proposed in [22]. It is a combination of SI and SC, which increases the DC-gain and reduces the higher switching voltage across the power switches.

switched inductor	N_s	N_L	N_C	K		$\eta\%$	
				up	down	up	down
[20]	9	2	5	$\frac{2(3+D)}{(1-D)}$	$\frac{D}{2}$	91.6	91.8
[21]	3	2	2	$\frac{(1+D)}{(1-D)}$	$\frac{D}{2-D}$	95.1	95.3
[22]	4	3	2	$\frac{2(1+D)}{(1-D)}$	$\frac{D}{4-3D}$	93.2	93.4

B. Isolated Bi-directional DC-DC Converters

The isolated bidirectional converter has the advantage of galvanic isolation. Whenever the fault occurs, it provides the isolation between the circuit's input and output sides. By adjusting the transformer's winding turn's-ratio, the transformer offers a wideband of voltage gain. These converters are more suitable for various applications due to their high voltage gain and isolation.

1) Flyback

In Flyback dc-dc converters [23-26], voltage conversion ratio can be adjusted by inductor turns ratio. However, it has the main advantages of galvanic isolation and good dynamic behavior. Conventional Flyback converter [23] has high switching stress across the power switches due to leakage flux of inductor. To overcome the high switching stress difficulty, the Flyback with snubber topologies are proposed in [24-26].

Flyback	N_s	N_L	N_C	$\eta\%$	
				up	down
[23]	4	-	2	82	82
[24]	4	-	4	87.2	87.2
[25]	12	-	1	92	92
[26]	4	-	4	91.3	87.3

RCD snubbers [25] are used for voltage clamping or to control the rate of rising voltage within a short duration.

Active clamped [24] is proposed to recover the energy stored in leakage inductance and reduces the voltage ripples. NCC-LCD [26] is presented for recycling the magnetizing energy, and CC-LCD [26] overcomes the circulating current. Here, two switching combinations of clamped approach is used to reduce the current stress.

2) Isolated CUK& SEPIC/ZETA Converter

The bidirectional isolated CUK converter [27] has one transformer, two switches, and one diode, a combination of three-magnetic components (two inductors & transformer). It eliminates the current ripples at input and output side inductors. It has other advantages of higher power density and lesser operating-frequency. The SEPIC/ZETA converter can operate in both zeta mode & SEPIC mode individually. Active clamping cell [28] is also included in isolated SEPIC/ZETA converter; it consists of a switch, clamping switch & clamping capacitor to minimize the leakage inductor energy.

CUK& SEPIC/ZETA	N_s	N_L	N_C	$\eta\%$	
				up	Down
[27]	5	-	2	85	85
[28]	5	-	4	89	89

3) Push-pull converter

Push-pull converter topology [29, 30] is different from the conventional isolated converter; In this topology, the transformer is used to control the output voltage. It stores the energy in one part of the cycle and supplies energy in the other part of the cycle. These can decrease the spikes in current of the transistors and extend the voltage conversion. Generally, push-pull converter suffers from ripples at input current; and it is rectified by using a filter circuit. This converter is more efficient at higher ratings. Push-pull converter topology [31] is used to reduce the switching components, besides the ZVS condition is also achieved.

Push-pull	N_s	N_L	N_C	$\eta\%$	
				up	Down
[29]	6	1	2	94.1	94.5
[30]	4	2	1	91	91
[31]	6	2	1	94	94

4) Forward Converter

The forward converter [33] supports achieving ZVS operation and high voltage gain with minimizes ripples in input current. The forward-bidirectional converter in [34], is used to extend the soft switching with proper design. In [32], a forward-bidirectional converter is altered with a clamping circuit to improve performance and to achieve zero voltage switching action.

Forward	N_s	N_L	N_C	$\eta\%$	
				up	Down
[32]	4	3	1	88	88
[33]	4	1	2	91	91
[34]	8	1	1	92.5	93

5) Dual Active Bridge (DAB)

DAB [35] is a popular topology widely used in high-power automotive system applications. But the amount of power transfer depends on the number of switches and is

controlled by PWM plus phase shift control. It has three stages of conversion based on structure. The first stage is DC-AC converting using a current-fed or voltage-fed full-bridge. In the second stage, the transformer helps to step-up the voltage and offers the galvanic-isolation. Sometimes, resonant-tank circuitry is used for ZVS/ZCS operation to enhance the performance. The third stage of conversion is AC-DC rectification through any voltage or current-fed full bridge. Dual Half-bridge converter [36] is having a half-bridge structure on either side of the high-frequency transformer. Generally, the dual half-bridge is used for lower power level applications; in this current-fed topology enables the continuous current waveform. And the interleaved half-bridge is used to boost up the voltage capacity and reducing the current stress. Half-bridge-full-bridge topology [37] is more suitable for UPS application; it has simpler control than PWM plus phase-shift like DAB controller. In [38], multiport DAB is proposed to charge the EV with ultra-fast mode, and also it supports the power in V2G mode. It monitors the battery state of charge (SoC) during off-load and peak load hours using an optimal algorithm. In addition to the above advantages, it also reduces the size.

DAB	N_s	N_L	N_C	$\eta\%$	
				up	down
[35]	8	2	2	97.8	97.8
[36]	4	1	4	93.4	93.4
[37]	7	-	2	95	95
[38]	4n	1	n	93.45	93.45

IV. SUMMARY

From the above-detailed literature review, the voltage gain ratio of various non-isolated Bi-directional dc-dc converter topologies is studied. The characteristics are plotted below under both buck and boost modes of operations. The different methodologies, features, merits, demerits, power densities, and efficiencies are summarized in Table 1. From this, the researcher can choose the most suitable converter based on their requirement and application.

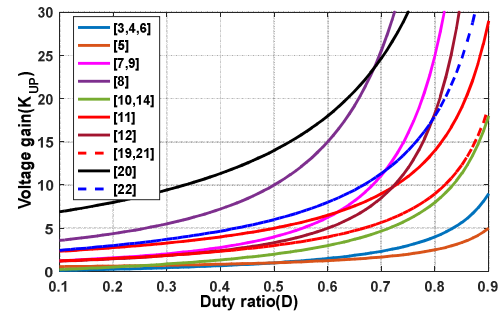


Fig-3.voltage gain vs duty ratio plot of non-isolated bidirectional dc-dc converters in step up mode

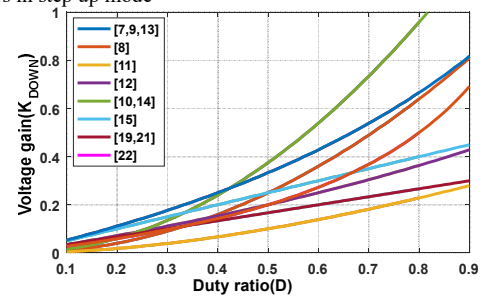


Fig-4.voltage gain vs duty ratio plot of non-isolated bidirectional dc-dc converters in step down mode

SUMMARY OF BIDIRECTIONAL DC-DC CONVERTER

Methodology	Features	Demerits	Efficiency	Power density	Component utilization& complexity	Gain
Flyback	1.Galvanic isolation without transformer 2.Good dynamic behaviour	1.High voltage stresses 2.Complex gate driving circuit	Low	Low	Less	-
Isolated cuk & Sepic/zeta	1.Eliminates current ripples in input & output current	1.Bulk in size	Low	High	more	-
Push-pull	1.Achieving soft switching 2.Less THD 3.Reliability in power transfer	1. Small Current ripples in the input side. 2. Cost and size increase with the increase in power rating.	High	Moderate	more	-
Forward	1.Achieving ZVS	1. Additional circuits are required to minimize voltage spikes 2. snubber is required	Moderate	Moderate	moderate	-
DAB	1.Achieving ZVS&ZCS, 2.integration of different voltage source 3. high power density	1.High input ripple current 2.Complicated control circuit	Moderate	High	More	-
Buck-boost	1.Simple in structure 2. fewer components	1.Produces negative output, which is not suitable for many applications 2.lower power transfer capability	Low	Low	Less	Low
Cuk&sepic-zeta	1.Eliminates current ripples in input & output current	1.High voltage stress 2.Limited voltage gain	Low	Moderate	More	Low
Switched capacitor	1.Good dynamic behaviour 2. low cost and lightweight circuits 3.decentvoltage gain with less components	1. Voltage gain mainly depends on the number of switched capacitors, leading to increased component count. 2.Lower range of discrete voltages	Low	High	Less	Moderate
Cascaded	1.high voltage gain 2.reduces the output current ripples in both modes	1.low efficiency 2. High voltage stress. 3.higher components	Low	Moderate	More	High
interleaved	1. It can maintain high efficiency while varying the different power levels. 2.No transformer for boosting 3.Active clamp circuit using converter	1. Voltage conversion ratio, 2. switching voltage stress is more 3.Control is complex	High	Low	More	Low
Multilevel	1.Modularity structure 2.High power capability	1. More number of components 2. Bulky, more weight and high cost.	Low	High	More	Low
Coupled	1. Coupled inductors high voltage boosting ability due to adjustable tum's ratio. 2. All switches can be adjusted to low voltage side which leads to reduction in switching losses.	1. effect of leakage inductance such as high voltage ripples. 2.Relatively bulky 3.Snubber circuits circuit is mandatory	High	Low	Less	High
Switched inductor	1. Extensive voltage boost ability. 2. Suitable for medium power applications due to less ripple in current.	1.Need more number of passive components 2.voltage gain mainly depends on the number of switched inductors, leading to increased component count and bulky	Moderate	Low	More	Moderate

V. CONCLUSION

This paper discusses about various non-isolated and isolated bidirectional DC-DC topologies. The detailed literature details of bidirectional dc-dc converter mainly focus on specific categories such as power density, cost, voltage gain, efficiency, and reliability. For better understanding, the same has been summarized in comparative Table 1. The merits, demerits, and performance of various topologies have been discussed in

detail to choose the optimal converter. The voltage conversion ratio of non –isolated converters are plotted under both step up and step down modes. In addition, the efficiencies of each converter are listed in the comparative tables. From this comprehensive analysis, one can thoroughly assess the performance and select the particular application's topology

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